

Testing mitochondrial markers for noninvasive genetic species identification in European mammals

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Received: 17 July 2013 / Accepted: 14 October 2013 / Published online: 27 October 2013
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Abstract While DNA-barcoding routines have largely facilitated molecular identification of animal species, there is no common strategy for species identifications based on noninvasively collected samples. We compared the performance of six commonly used mitochondrial fragments for identification success of low-quality hair and scat samples from 15 European mammals. The standard cytochrome oxidase subunit 1 barcoding fragment performed worst in the test, with only three successful species identifications out of 32 samples. In contrast, three short fragments (16S and 12S rDNA, cytochrome b) led to reliable species identifications for >80 % of all tested samples. Success rates also differed among feeding types.

Keywords Noninvasive sampling · Species discrimination · Mammalia · Genetic monitoring · Wildlife forensics

Genetic monitoring based on the use of noninvasively collected samples has become an important tool in monitoring programmes and field studies of endangered mammals (Schwartz et al. 2007). While most respective studies focus on single target species (e.g., Perez et al. 2009; Karamanlidis et al. 2012) or, for example, carnivore

communities (Ruell and Crooks 2007; Fernandes et al. 2008; Gompper et al. 2006), suitable genetic identification systems for entire mammal communities have not gained significant interest so far. In order to fill this gap we tested six primer pairs identified from an extensive literature survey that are commonly used for species identification on hair and scat samples from various medium-sized and large European mammals.

Thirty-two samples (16 scat and 16 hair samples) of 15 species were taken from our internal data base of noninvasively collected mammal samples. We explicitly preferred samples with presumably low DNA quality, for instance due to progressed sample age or small amount of available material. The samples were derived from German wildlife parks (European wildcat, Eurasian lynx, brown bear, red deer, moufflon, goat) or field samples collected across Germany and Austria (European wildcat, Eurasian otter, raccoon, red fox, Eurasian beaver, horse, wild boar, red deer, roe deer, European hare) (Table 1).

Samples were extracted in a laboratory dedicated to the pre-PCR processing of noninvasively collected materials using the commonly used Qiagen Investigator Kit (hairs) or the Qiagen Stool Kit (scats) following the recommendations of the manufacturer with slight modifications regarding dilution volumes. Mitochondrial (mt) DNA fragments were amplified from all 32 samples using six different primer pairs found in the literature (for/rev): 1. (CrL, control region) Thr-L15926 5'CAATCCCCGGT CTTGTAAACC/DL-H16340 5'CCTGAAGTAGGAACAGATG (Cheney 1995), (Vila et al. 1999); 2. (CrS, control region) L15995 5'CTCCACTATCAGCACCCAAAG/H16498 5'CCTGAAGTAAGAACCAGATG (Pun et al. 2009); 3. (COI, cytochrome oxidase 1) LCO1490 5'GGTCAACA AATCATAAAGATATTGG/HCO2198 5'TAAACTTCAG GGTGACCAAAAAATCA (Folmer et al. 1994); 4. (CytB

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Table 1 Species identification of 32 hair and scat samples from 15 mammal species with six primer combinations (see text for details). Shown are sample number (No.), species names, sampling date, and sampling location. Black dots indicate successful species identification via BLAST search. CRL values show the maximum continuous readable sequence length in the sequence trace file

Sample information				Detection success													
No.	Species	Type	Date	Location	CrL		CrS		COI		CytB		12S		16S		
					ID	CRL	ID	CRL	ID	CRL	ID	CRL	ID	CRL	ID	CRL	
Fs01	<i>Felis silvestris</i>	Hair	06/03/2010	Stadtwald Beverungen		0		0		0		^a	175	^a	145	^a	169
Fs02	<i>Felis silvestris</i>	Scat	08/01/2008	Opel-Zoo Kronberg	^a	215	^a	338		0		^a	175	^a	139	^a	157
Fs03	<i>Felis silvestris</i>	Scat	08/01/2008	Opel-Zoo Kronberg	^a	252	^a	499		0		^a	175	^a	155	^a	181
Ly01	<i>Lynx lynx</i>	Hair	12/18/2009	Wildpark Knüll		18		71		0			175		145		183
Ly02	<i>Lynx lynx</i>	Scat	12/09/2009	Wildpark Lüneb. Heide		36		110		1			205		157		176
Ly03	<i>Lynx lynx</i>	Scat	12/09/2009	Wildpark Lüneb. Heide		0		573		198			147		147		162
Lu01	<i>Lutra lutra</i>	Scat	06/14/2010	Lausitz, Saxony	^a	435		0		0			0		60		185
Cl01	<i>Canis lupus</i>	Hair	02/02/2007	Lausitz, Saxony	^b	251	^b	328		0		^b	157	^b	167	^b	184
Cl02	<i>Canis lupus</i>	Scat	12/29/2006	Lausitz, Saxony	^b	298	^b	234		0		^b	175	^b	145		20
Cl03	<i>Canis lupus</i>	Scat	02/07/2009	Lausitz, Saxony	^b	384		Not analysed		0		^b	175	Not analysed	Not analysed	Not analysed	Not analysed
Vv01	<i>Vulpes vulpes</i>	Scat	03/16/2009	Spessart mountains		0		301		39			178		152		176
Vv02	<i>Vulpes vulpes</i>	Scat	03/16/2009	Spessart mountains		0		0		0			230		136		40
Pt01	<i>Procyon lotor</i>	Hair	03/16/2009	Spessart mountains		6		Not analysed		0			169	Not analysed	Not analysed	Not analysed	Not analysed
Pt02	<i>Procyon lotor</i>	Hair	03/16/2009	Spessart mountains		0		0		0			165		86		108
Ua01	<i>Ursus arctos</i>	Hair	07/22/2009	Wildpark Westerwald		0		15		0			0		0		154
Ua02	<i>Ursus arctos</i>	Hair	07/22/2009	Wildpark Westerwald		0		0		0			0		0		0
Ua03	<i>Ursus arctos</i>	Scat	08/12/2009	Bärenpark Worbis		18		9		0			174		143		72
Ua04	<i>Ursus arctos</i>	Scat	08/12/2009	Bärenpark Worbis		0		359		0			156		141		35
Cf01	<i>Castor fiber</i>	Hair	11/21/2009	Ansbach, Bavaria		516		447		158			175		159		166
Cf02	<i>Castor fiber</i>	Hair	12/01/2010	Reimlingen, Bavaria		29		449		0			176		144		175
Le01	<i>Lepus europaeus</i>	Hair	02/25/2009	Spessart mountains		420		Not analysed		0			174	Not analysed	Not analysed	Not analysed	Not analysed
Le02	<i>Lepus europaeus</i>	Hair	03/11/2009	Spessart mountains		337		Not analysed		0			172	Not analysed	Not analysed	Not analysed	Not analysed
Ec01	<i>Equus caballus</i>	Hair	10/02/2010	Taunus mountains		355		318		0			0		164		184
Ec02	<i>Equus caballus</i>	Scat	10/02/2010	Taunus mountains		0		9		0			0		146		161
Ss01	<i>Sus scrofa</i>	Hair	03/11/2009	Spessart mountains		357		401		0			177		147		173
Ss02	<i>Sus scrofa</i>	Scat	02/10/2010	Lungau, Austria		0		317		0			173		0		174
Cc01	<i>Capreolus capreolus</i>	Hair	09/30/2010	Kellerwald mountains		383		Not analysed		0			172	Not analysed	Not analysed	Not analysed	Not analysed
Ce01	<i>Cervus elaphus</i>	Hair	03/04/2009	Spessart mountains		106		Not analysed		72			26	Not analysed	Not analysed	Not analysed	Not analysed
Ce02	<i>Cervus elaphus</i>	Scat	10/21/2010	Spessart Wildpark		18		Not analysed					143	Not analysed	Not analysed	Not analysed	Not analysed
Ce03	<i>Cervus elaphus</i>	Scat	10/21/2010	Spessart Wildpark		0		0		0			0		164		184
Oo01	<i>Ovis orientalis</i>	Scat	10/21/2010	Spessart Wildpark		21		475		8			386		141		178
Ca01	<i>Capra aegagrus</i>	Hair	10/21/2010	Spessart Wildpark		59		109		0			176		162		184

Table 1 continued

Sample information					Detection success											
No.	Species	Type	Date	Location	CrL		CrS		COI		CytB		12S		16S	
					ID	CRL	ID	CRL	ID	CRL	ID	CRL	ID	CRL	ID	CRL
Average					15/32	146	16/25	210	3/32	17	26/32	148	21/25	126	21/25	143
% successful species identifications					47 %		64 %		9 %		81 %		84 %		84 %	

^a Differentiation among European wildcats and domestic cats is blurred due to hybridisation; we do not differentiate among both forms here

^b Only control region may differentiate among wolves and domestic dogs when using mitochondrial DNA. We do not consider this differentiation here

cytochrome B) CanidC1 5'/AATGACCAACATTCGAAA (Paxinos et al. 1997)/HCarn200 5'/ATTCAGCCRTARTTA ACGTC (Bidlack et al. 2007); 5. (12S) L1085 5'/CCCAA ACTGGGATTAGATACCC/H1259 5'/GTTTGCTGAAG ATGGCGGTA (Kitano et al. 2007); 6. (16S) L2513 5'/GC CTGTTTACCAAAAACATCAC/H2714 5'/CTCCATAGG GTCTTCTCGTCTT (Kitano et al. 2007). Due to limitation of available DNA, however, three primer pairs were not tested for seven samples (Table 1).

We used a standard PCR protocol with 15 µl reaction volumes consisting of 3 µl undiluted DNA template, 0.66 U Mg-free TAQ DNA polymerase (New England Biolabs), 1 × PCR buffer (New England Biolabs), 3 mM MgCl₂, and 0.3 µM of each primer. Amplification was performed with 5 min initial denaturation at 94 °C, 40 cycles of 45 s at 94 °C, 30 s at 54 °C, 30 s at 72 °C, and a 10 min final elongation at 72 °C. Purified amplicons were run on a 3730 DNA analyzer (Applied Biosystems). Sequences were processed manually and quality checked using sequence scanner software v1.0 (Applied Biosystems). All sequences were blasted using default settings in the National Center for Biotechnology Information (NCBI) GenBank.

Safe species determination was possible in most cases with readable continuous fragment length (CRL) of >50 bp (Table 1). While all primers generated at least some readable fragments allowing for correct species identification, amplification success and resulting rates of successful species determination differed considerably among markers. Only three samples could be successfully assigned to the correct species with the 658 bp 'barcoding fragment' (COI), indicating its limited suitability for noninvasively collected samples due to its considerable size (Table 1). While the control region is a frequently targeted region in numerous studies on genetic diversity in mammals due to its high mutation rates, the two tested, size variable (~370–800 bp, depending on the respective species) fragments provided intermediate success rates (47 and 62 % for CrL and CRS, respectively). In contrast, use of any of the three remaining fragments (CytB, 12S, and 16S) resulted in high rates of successful species identifications of >80 %. With length of 215 (12S), 234 (CytB) and 244 bp (16S) these fragments are well in the range of markers commonly used for samples of low DNA quality (Taberlet et al. 1999).

To a certain extent, all primers amplified across taxa and no clear trends were observed here based on this rather small sample. However, predominantly carnivorous species showed higher success rates than herbivores when using scats (68 % compared to 48 %, respectively, Table 2), which can be explained by the presence of inhibitory substances in plant tissues. Interestingly, the opposite picture was revealed for hair samples. Overall

Table 2 Species identification success in hair and scat samples of carnivores (Fs, Ly, Lu, Cl, Vv, Pl, Ua; see Table 1) and herbivores (Cf, Le, Ec, Ss, Cc, Ce, Oo, Ca). Note that the distinction among both feeding types is not rigorous as we included partially herbivorous carnivores, such as Ua, Pl, and Vv into the carnivore guild

Sample type	Scat				Hair			
	n	sp.	ID	%	n	sp.	ID	%
Carnivores	11	6	41/60	68	7	5	17/39	44
Herbivores	5	4	13/27	48	9	7	32/42	76
Total	16	10	54/87	62	16	12	49/81	60

we show the general suitability of short mtDNA fragments for simple species determination of European mammals based on noninvasively collected samples. Due to the overall high availability of cytochrome B sequence data in mammals, our tested CytB primer pair might be generally preferred when one has to rely on publicly available sequence database information for species determination. However, we also stress the importance for considering potential limitations of using single mtDNA markers arising from current or past hybridisation, lineage sorting or high intraspecific and cryptic variation.

Acknowledgments Several wildlife parks provided kind sampling assistance. Sequence reaction were performed in the BiK-F Laboratory Centre, Frankfurt. This project was funded by the Leibniz-Association in the context of the “Joint Initiative for Research and Innovation” (SAW-Pakt).

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